KCA UNIVERSITY

BSD 3204: MACHINE LEARNING — C.A.T 2

MARCH 2026

Question a) Four Real-World ML Applications in Kenya (4 Marks)

1. Mobile Banking and Fraud Detection — M-Pesa and other mobile money platforms use ML algorithms to detect unusual transaction patterns and flag potential fraud in real time. This protects millions of Kenyans from financial fraud.

2. Agricultural Advisory Systems — Platforms like Safaricom's Digifarm use ML to analyze soil data, weather patterns, and satellite images to give farmers crop recommendations tailored to their specific regions, helping improve yields.

3. Healthcare Diagnostics — Hospitals and health startups are using ML to help diagnose diseases like tuberculosis and malaria by analyzing medical images and patient records, improving accuracy especially in areas with few specialists.

4. Credit Scoring for the Unbanked — Fintech companies like Tala and Branch use ML to assess creditworthiness of people without formal credit history by analyzing mobile phone usage patterns, enabling them to access loans.

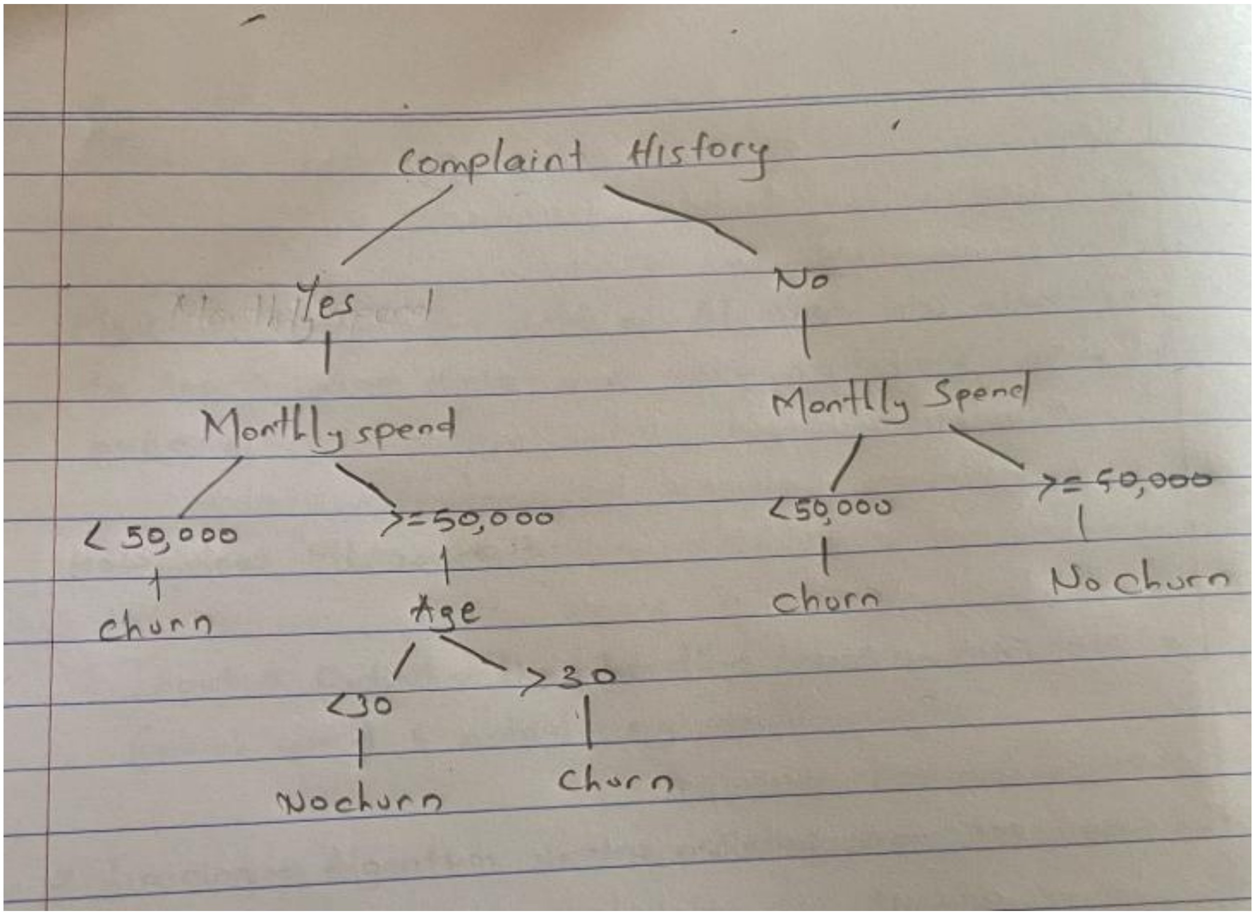
Question b) Two Ethical Risks of Unregulated ML in Kenya's Public Sector (4 Marks)

1. Bias and Discrimination — ML systems trained on historical data can perpetuate existing inequalities. For example, if a government welfare targeting system is trained on biased data, it may unfairly exclude certain ethnic groups or regions from receiving benefits. In Kenya's diverse setting, this could worsen social inequalities if left unchecked.

2. Lack of Accountability and Transparency — When government decisions (such as loan approvals, police profiling, or student placement) are made by ML models, it becomes hard for citizens to challenge unfair outcomes. Without regulation, there is no obligation to explain how a model arrived at a decision, which undermines democratic accountability and citizens' rights.

Question c) Decision Tree for Customer Churn Prediction (4 Marks)

The decision tree below predicts whether a customer will churn (leave) based on Age, Monthly Spend, and Complaint History:



Explanation: Customers with a complaint history and low spend are likely to churn. Younger customers with no complaints but low spend may still churn. High-spending customers with no complaints are generally retained.

Question d) Naive Bayesian Classifier for Student Level Prediction (4 Marks)

The task is to classify a student into one of several English proficiency levels (e.g., Beginner, Intermediate, Advanced) using three features: Age, First Language (L1), and Years of Prior English Study.

Step 1 — Training (Parameter Estimation):

From historical data, we estimate the following probabilities:

Prior Probability of each class:

P(Level = C) = (Number of students in class C) / (Total students)

Likelihood of each feature given a class:

For categorical features like L1:

P(L1 = x | Level = C) = (Count of students in class C with L1 = x) / (Total students in class C)

For continuous features like Age and Years of Study, we assume a Gaussian (Normal) distribution:

$P(\text{feature} = x \mid \text{Level} = C) = (1 / \text{sqrt}(2\pi \sigma^2)) \times \exp(-(x - \mu)^2 / 2\sigma^2)$

where μ = mean and σ^2 = variance of the feature within class C

Step 2 — Classification:

For a new student with features (Age=a, L1=l, YearsStudy=y), we compute the posterior for each class C:

$P(C \mid \text{Age, L1, YearsStudy}) \propto P(C) \times P(\text{Age} \mid C) \times P(\text{L1} \mid C) \times P(\text{YearsStudy} \mid C)$

We assign the student to the class with the highest posterior probability:

Predicted Level = argmax_C [P(C) × P(Age|C) × P(L1|C) × P(YearsStudy|C)]

The 'Naive' assumption is that Age, L1, and YearsStudy are independent of each other given the class — this simplifies the computation and works well in practice even when the assumption is not perfectly true.

Question e) Naïve Bayes Spam Classification (8 Marks)

We need to classify the email "offer discount click" as Spam or Not Spam.

Email ID	Offer	Click	Discount	Urgent	Spam
1	1	0	1	0	Yes (1)
2	0	1	1	1	Yes (1)
3	1	0	0	1	No (0)
4	0	1	1	0	No (0)

Step 1 — Prior Probabilities:

Total emails = 4

Spam emails = 2 (emails 1 and 2)

Not Spam emails = 2 (emails 3 and 4)

P(Spam) = 2/4 = 0.5

P(Not Spam) = 2/4 = 0.5

Step 2 — Likelihoods (using frequency from training data):

For each word, we count how many spam and not-spam emails contain it:

Word	Count in Spam (out of 2)	P(word=1|Spam)	Count in Not Spam (out of 2)	P(word=1|Not Spam)
offer	1 (email 1)	1/2 = 0.5	1 (email 3)	1/2 = 0.5
click	1 (email 2)	1/2 = 0.5	1 (email 4)	1/2 = 0.5
discount	2 (emails 1,2)	2/2 = 1.0	1 (email 4)	1/2 = 0.5
urgent	1 (email 2)	1/2 = 0.5	1 (email 3)	1/2 = 0.5

Step 3 — Classify "offer discount click":

We only use the three words present: offer=1, discount=1, click=1

Score for Spam:

$$P(\text{Spam}) \times P(\text{offer}=1|\text{Spam}) \times P(\text{discount}=1|\text{Spam}) \times P(\text{click}=1|\text{Spam})$$

$$= 0.5 \times 0.5 \times 1.0 \times 0.5$$

$$= 0.5 \times 0.25 = 0.125$$

Score for Not Spam:

$$P(\text{Not Spam}) \times P(\text{offer}=1|\text{Not Spam}) \times P(\text{discount}=1|\text{Not Spam}) \times P(\text{click}=1|\text{Not Spam})$$

$$= 0.5 \times 0.5 \times 0.5 \times 0.5$$

$$= 0.5 \times 0.125 = 0.0625$$

Step 4 — Decision:

$P(\text{Spam} \mid \text{email}) = 0.125$

$P(\text{Not Spam} \mid \text{email}) = 0.0625$

Since $0.125 > 0.0625$, the email "offer discount click" is classified as → SPAM ✓

Question f) Confusion Matrix Calculations (6 Marks)

	Predicted Spam	**Predicted Not Spam**
Actual Spam	TP = 45	FN = 5
Actual Not Spam	FP = 10	TN = 40

i. Precision (2 Marks):

Precision = TP / (TP + FP)

Precision = 45 / (45 + 10)

Precision = 45 / 55

Precision ≈ 0.818 or 81.8%

This means that when the model predicts an email is spam, it is correct about 82% of the time.

ii. Recall (2 Marks):

Recall = TP / (TP + FN)

Recall = 45 / (45 + 5)

Recall = 45 / 50

Recall = 0.9 or 90%

This means the model correctly identifies 90% of all actual spam emails.

iii. Accuracy (2 Marks):

Accuracy = (TP + TN) / (TP + TN + FP + FN)

Accuracy = (45 + 40) / (45 + 40 + 10 + 5)

Accuracy = 85 / 100

Accuracy = 0.85 or 85%

This means the model correctly classifies 85% of all emails overall.